

Technology News

Robotic help for partially paralysed

A new robotic device developed by Argo Medical Technologies in Haifa, Israel, would soon help partially paralysed people walk again. Christened 'ReWalk,' the robot uses a 'smart software' that understands what the user is intending to do and translates that into taking a step or climbing a stair. ReWalk helps paraplegics stand, walk and even climb stairs.

ReWalk is basically a wearable,



ReWalk robotic suit for partially paralysed

motorised quasi robotic suit. Partially concealable under clothing, it provides user-initiated mobility—leveraging advanced motion sensors, sophisticated robotic control algorithms, on-board computers, real-time software, actuation motors, tailored rechargeable batteries and composite materials.

Parking space finder for cars

Engineers at Rutgers University, New Jersey, have combined ultrasonic sensors, GPS receivers and cellular data networks to create a low-cost, highly effective method to find the nearest available parking space. The device will help motorists take decisions beforehand, as it has a Web-based interface. It will show roads that have congestion, so travellers can choose a different route for travel.

The sensor measures the distance to the next obstacle at the side of car. And if a car is driving in the right-most lane, then if cars are parked at the roadside, the sensor will register a small distance. If there is no car parked, it will register a larger distance, explains Marco Gruteser, assistant professor at Rutgers University.

Mission to the sun

Man has visited and even studied the moon, but the blinding atmosphere of the sun has always remained a mystery. That is about

to change soon. NASA has launched a \$808-million Solar Dynamics Observatory, from Florida's Cape Canaveral Air Force Station, to know more about our fiery neighbour and secrets it may harbour. The high-tech spacecraft has three separate instruments that will work together to study how solar activity gets its start. The results could help us to know more about the sun's wide variability, which has the potential to affect the earth's climate.

Clothes to power handheld gadgets

Scientists at University of California, Berkeley, have created energy-scavenging nanofibres that could one day be woven into clothing and textiles. These nano-sized generators have piezoelectric properties that allow them to convert into electricity the energy created through mechanical stress, stretches and twists. So they could be used to create clothing that can power handheld electronics through ordinary body movements. Because the nanofibres are made from organic polyvinylidene fluoride, these are flexible and relatively easy and cheap to manufacture. The tiny nanogenerators have diameters as small as 500 nanometres—about 100 times thinner than a human hair.

The researchers repeatedly tugged and tweaked the nanofibres, generating electrical outputs ranging from 5 to 30 mV and 0.5 to 3 nanoamperes. No degradation was observed after stretching and releasing the nanofibres for 100 minutes at a frequency of 0.5 hertz.

Circuit that powers itself

Scientists at the University of Pennsylvania have created the world's first photovoltaic circuit that can power itself from sunshine. This circuit could eventually lead to self-powered touchscreen devices.